

Khai Xuen Tang

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WORK EXPERIENCES

Software Developer Intern

01/2025 – 09/2025

Alpha Red Solutions

- Developed an admin dashboard featuring full CRUD capabilities for managing orders, invoices, hotels, and rooms, as well as handling corporate registrations and markup configurations.
- Leveraged Vue 3's built-in directives and Reactivity API to implement conditional rendering in entry forms and dashboards based on user input or user privileges, while asynchronous data fetching via REST APIs was handled using Watchers.
- Implemented strict type checking using TypeScript to prevent application failures caused by type mismatches and ensure seamless integration of tasks across team members.
- Collaborated with senior developers, backend teams, product managers, and Business Analysts in an Agile-based SDLC to understand user requirements and convert them into working features.

Part-Time Software Developer

09/2024 – 11/2024

Way To India

- Built an online travel booking platform for hotel and transportation ticket bookings in India using Next.js, TypeScript, and Tailwind CSS.
- Utilized Chrome DevTools for debugging, responsiveness testing, and performance analysis to quickly identify and resolve issues while enhancing the user experience.
- Collaborated with team members using version control tools such as Git to track changes and streamline development, integration, and deployment workflows.

EDUCATION

Software Design with Artificial Intelligence for Cloud Computing

09/2022 | Athlone, Ireland

Technological University of the Shannon, Midlands

Graduating with First-Class Honours and on the Dean's List for two consecutive years.

- Courses: Software Development, Machine Learning, Distributed Systems, Security, Networks, Databases, Data Visualization.

PROJECT EXPERIENCES

Blogging Platform

React.js, Tailwind CSS, Express.js, MongoDB, Firebase

- Developed a blogging platform with user authentication, authorization, blog management, search functionality, pagination, third-party API integration and user interactions to provide a modern blogging experience for users.
- Implemented security measures such as user authentication, configuring CORS, storing access and refresh tokens in HTTP-only cookies, sanitizing blogs and images submitted, and applying rate limiting to all APIs to protect the website.
- Designed MongoDB schemas with well-defined relationships and indexed key fields to improve query performance.

Real-time ray tracing software

WebGPU

- Designed and developed a 3D rendering software that supports both rasterization and ray tracing, dynamic shading and GLB model upload.
- Utilized Vite as a build tool to set up development environment and code bundler to enhance development speed and build optimized static assets for production.
- Developed WGSL shaders to utilize GPU's compute power to perform compute intensive tasks such as ray geometry intersection and shading.

Abalone age prediction machine learning model

Python · scikit-learn · Pandas · Matplotlib

- Utilized Linear Regression to predict abalone age from biological features, achieving an R^2 score of 0.54 after preprocessing the dataset through outlier removal and categorical feature encoding.
- Leveraged Matplotlib to identify outliers and visualize feature correlations, enabling deeper understanding of the dataset and feature relationships.
- Evaluated model performance using k-fold cross-validation and measured prediction error using RMSE and MAE.

SKILLS

Languages: Javascript, Python, Java, C++

Backend: Express.js, Springboot, PHP, JPA

Frontend: HTML, Javascript, Tailwind, React.js, Vue.js, Typescript, CSS, Next.js

DevOps: Git, Docker

Database: MySQL, MongoDB, Firebase